

DSA0602 – DATA HANDLING AND VISUALIZATIONS

ACTIVITY

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ROLL NO.06

SESSION 1 — ACTIVITY (20 MIN) • STUDENT HANDOUT (See End of Session 1 for more different data)

Decode the Chart

Name: _____ Roll No: _____

For each chart below, answer: What is it trying to tell you? Do you trust it? Why or why not?

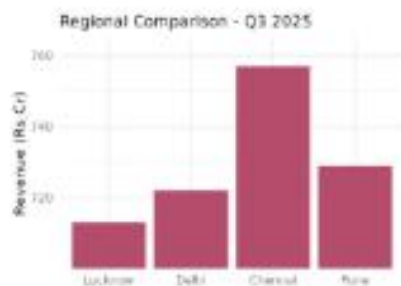


Chart 1

App Usage by Category (12 categories)



Chart 2

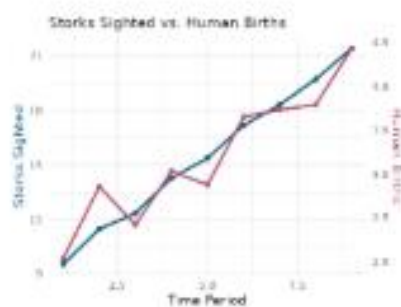


Chart 3

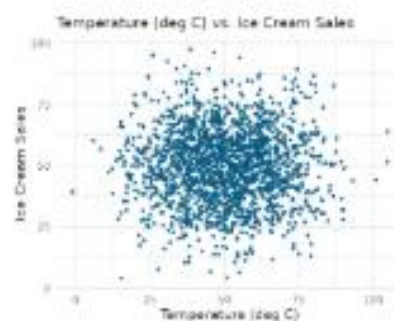


Chart 4

Chart 1 – Truncated Y-axis

What is it trying to tell you?

The chart compares the **population of different regions** such as Lucknow, Pune, Chennai, and other cities. It is intended to show which region has a higher or lower population.

Do you trust it? Why or why not?

No, not completely. The Y-axis does not start from zero. Because of this, the difference between the bars looks much larger than the actual difference in population. A viewer may think that one region has a much higher population than another.

Main issues:

- The Y-axis is truncated.
- Small differences appear visually large.
- The bar lengths can give a misleading impression.

- It makes comparison between regions less accurate.

Better way:

The Y-axis should start from **zero**. This allows the length of each bar to represent the actual difference in population.

Chart 2 – Too Many Categories in Pie Chart

What is it trying to tell you?

The chart shows the **distribution of app usage among different categories**. Each slice represents one category and its size represents its share of the total.

Do you trust it? Why or why not?

Not completely. The chart contains too many slices, making it difficult to compare their sizes. Some slices are very similar, so it is difficult to identify the exact differences between categories.

Main issues:

- Too many categories are included.
- Similar-sized slices are difficult to compare.
- Different colors make the chart harder to follow.
- It is difficult to identify the largest and smallest categories quickly.

Better way:

A **sorted bar chart** would be more effective because the categories can be arranged from highest to lowest, making comparison easier.

Chart 3 – Dual-axis / False Correlation

What is it trying to tell you?

The chart compares **two different variables**, such as storks sighted and human births. The two lines appear to move together, suggesting that the variables may have a relationship.

Do you trust it? Why or why not?

No. The chart uses two separate Y-axes with different scales. These scales can make two unrelated variables appear to follow the same pattern. Therefore, the visual similarity of the lines does not prove that one variable affects the other.

Main issues:

- Two different Y-axis scales are used.
- The lines appear related because of the chosen scales.
- Visual correlation does not necessarily mean actual correlation.
- It may lead the viewer to a wrong conclusion.

Better way:

Use **separate charts** for the two variables or use a **scatter plot and correlation coefficient** to properly investigate their relationship.

Chart 4 – Overplotting

What is it trying to tell you?

The chart shows the relationship between **temperature and ice cream sales** using a scatter plot. Each point represents an individual observation.

Do you trust it? Why or why not?

Not completely. There are many points plotted close together and overlapping. This makes it difficult to see the actual distribution and determine the strength or direction of the relationship.

Main issues:

- Too many points overlap.
- Dense areas appear as large blobs.
- Individual observations are difficult to identify.
- Outliers may be hidden.
- The overall pattern is difficult to see clearly.

Better way:

Use **transparent points (alpha)**, a **hexbin plot**, or a **2D histogram**. These methods can show where observations are concentrated more clearly.
